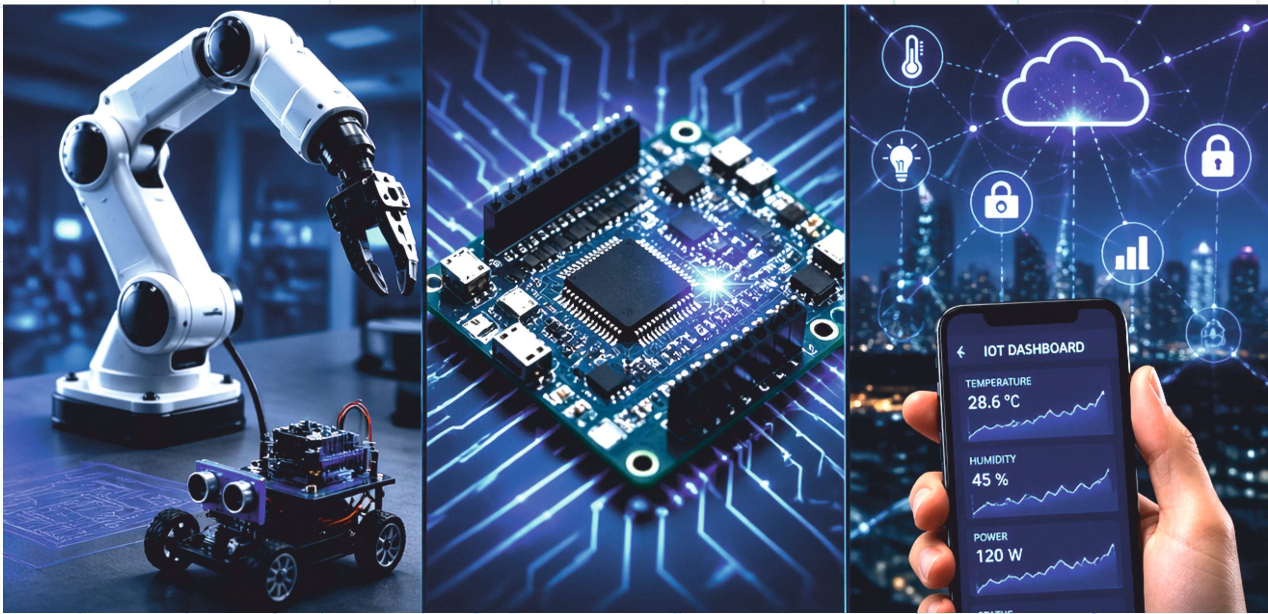
ONE MONTH INTERNSHIP PROGRAM

on Robotics, Embedded Systems & IoT



PROGRAM OVERVIEW

This 1-month intensive internship program is designed to provide students with practical exposure, industry-relevant skills, and hands-on experience in emerging Robotics, Embedded Systems & IoT technologies.

The program focuses on learning by doing, where students will work on real hardware, live demonstrations, and guided project development.

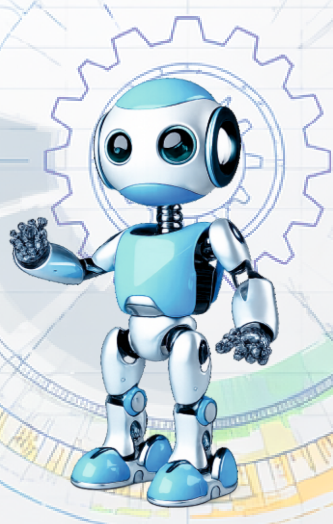
By the end of the program, students complete their working project, understand industry best practices, and develop the confidence to prototype, implement, and present technology solutions relevant to modern automation and smart systems.

This internship not only enhances technical competence but also nurtures creativity, teamwork, documentation skills, and professional readiness, preparing for careers and higher studies in cutting-edge technology domains.

Program Duration

Duration: 1 Month (4 Weeks)

Batch Size: 60 Students



Module 1.

Fundamentals & Live Demonstrations

A Foundational Exploration through Multi-Platform Live Demonstrations

Introduction to Robotics

- Basics of Robotics and its importance in modern industries
- Demonstration of real robotic systems:
  - 9 DOF Biped Robot
  - Spider Robot
  - Mini Robot
  - Droid Robot
  - Otto Bot
  - Robo-Dog
  - Smart AI Robotics Car
  - Robotic Arm
- Applications in industries like manufacturing, healthcare, defence, and automation

Module 2.

Introduction to Artificial Intelligence in Robotics

- Concept of AI and its integration with robotics
- Real-time demonstration of a Humanoid Robot
- Understanding intelligent behaviour and automation
- Different sub-sets of AI Model
- AI-ML-DL-GPT

Module 3.

Introduction to Drone Technology

- Basics of drones and their working principles
- Understanding the Drone dynamics
- Types of drones:
  - F330 Drone
  - F450 Drone
  - Q250 Drone
  - S500 Drone
  - Defence-purpose drones
- Applications in surveillance, agriculture, mapping, and defence
- Live working model demonstrations

Module 4.

Introduction to IoT (Internet of Things)

- Fundamentals of IoT and smart systems
- Applications in daily life (Smart Home, Automation, Industry)
- Demonstration of real IoT-based working models

Module 5.

Arduino & Embedded Programming

Arduino IDE & Basic Hardware Interfacing.

Understanding the Atmega Microcontroller architecture, fan-in, fan-out, timing control, PWM modulation, and peripheral interfacing.

- Introduction to Arduino IDE
- Writing and uploading programs
- Hands-on activities:

- LED Blinking
- Delay Functions
- LED Intensity Control
- Fading Effects
- Multiple Hardware Control

Module 6.

Sensor Data Processing

Real-time data analysis, embedded system Data synthesis and implementation, controlling Fan-Outs.

- Understanding Analog vs. Digital Signals
- Hands-on activities:
  - Analog Data Reading
  - Digital Data Reading
  - Serial Monitor Output
- Working with sensors and real-time data visualization

Module 7.

PWM & Motor Control

- Concept of PWM (Pulse Width Modulation), creating a variable threshold signal, and H-bridge-based bidirectional DC motor control.
- Hands-on activity:
  - Motor speed control
  - Direction control
  - Speed variation using programming

Module 8.

Robotics Project Development

Password-Based Security System. An embedded data-based authentication system for a secured access.

- Concept of embedded security systems
- Hands-on development using:
  - Keypad
  - Microcontroller
  - Output devices
- Real-world applications (Door locks, secure access systems)

Module 9.

Line Following Robot

- Working principle of Infra-Red sensors. Data types of the IR sensor. Using IR-OUTPUT in analog and digital Modes.
- Robot navigation logic
- Hands-on development of:
  - Line Following Robot
- Applications in industrial automation

Module 10.

Obstacle Avoiding Robot

- Introduction to ultrasonic (sonar) sensors. Sonar technology, the concept of echo & trigger pin.
- Motor driver (L298N) working
- Hands-on development using:

- Atmel Microcontroller
- Ultrasonic Sensor
- Motor Driver

- Autonomous navigation concepts

Module 11.

Smart Systems & IoT Projects

Understanding the core concept of Internet of Things, Bluetooth technology, and Device Integration.

Bluetooth Controlled Smart Home Automation

- Communication using the HC-05 Bluetooth module
- RX-TX serial communication
- Controlling devices via mobile app
- Hands-on development of:
  - Smart Home Automation System

Module12.

Smart Temperature-Controlled Fan System

Variable threshold signals, setting-up parameters for variable functions.

- Working of DHT11 sensor
- Temperature & humidity-based automation
- Hands-on development:
  - Automatic Smart Fan System
- Real-life IoT application implementation

Module 13.

Advanced IoT Live Monitoring & Alert System

Understanding the core concept of Internet of Things, IoT architecture, and Device Integration.

- Hands-on development using:
  - MQ3 Sensor (Gas/Alcohol Detection)
  - DHT11 Sensor (Temperature & Humidity)
  - ESP Microcontroller (Wi-Fi-enabled)
- Cloud integration using Thing Speak IoT Platform
- Features:
  - Real-time data monitoring
  - Live alert system (threshold-based triggering)
  - Data visualization on dashboard (graphs & analytics)
- Concepts covered:
  - IoT architecture (Device → Cloud → User Interface)
  - API integration
  - Remote monitoring systems

Learning Outcomes

By the end of the internship, students will:

- Understand core concepts of Robotics, AI, IoT, and Drones
- Gain practical experience with Arduino and embedded systems
- Build and program real-world robotic projects
- Develop problem-solving and technical skills
- Get exposure to industry-level technologies

Certification

- Internship Completion Certificate
- Project Participation Certificate

Who Can Join?

- Diploma / B.Tech / M.Tech Students
- BCA / MCA / BSc IT Students
- Any technical background learner interested in Robotics & AI

Workshop Methodology

1. On-screen explanation of components, circuits, and programs through a projector.
2. Trainers will be assigned to help participants through real-time development.
3. Workshop Execution, Students' Interaction, Q&A session.

Enrollment & Payment Guidelines

1. Enrollment confirmation (by institution/student) must be completed at least 10 days prior to the program commencement.
2. Payment is required in advance to confirm participation and reserve the slot.
3. Seats are limited and will be allotted on a first-come, first-served basis.

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